

Danny Rollo

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RESEARCH INTERESTS

Deep learning for medical imaging and healthcare; efficient architectures for low-data regimes; representation learning and recommender systems; imitation learning for robotic manipulation; agentic LLM systems.

EDUCATION

Northeastern University

M.S. in Computer Science

Boston, MA

Sept 2024 – Expected May 2026

B.S. in Computer Science & Mathematics

Sept 2021 – May 2025

- *Graduate coursework:* Applied Deep Learning, Deep Learning, Machine Learning, Algorithms, Empirical Research Methods, Web Development, Computer Graphics
- *Undergraduate coursework:* AI4Impact, Matrix Methods in Machine Learning & Data Analysis, Systems Security, Object-Oriented Design, Statistics & Stochastic Processes

PUBLICATIONS

Peer-Reviewed Conference

1. **Rollo, D.** “Lightweight CNN Encoders for Nucleus Segmentation.” *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2026. **Accepted — poster.**

A controlled encoder ablation showing a 34M-parameter VGG U-Net matches MedT — a transformer purpose-built for small medical datasets — at a $\sim 24\times$ smaller parameter budget, and outperforms pretrained Swin-T by 4.9 Dice points with $2.6\times$ less memory and $1.8\times$ lower inference latency.

In Preparation

2. **Rollo, D., et al.** “Two-Tower Recommendation with Cold Start Ablation.” *Manuscript in preparation.*

A 12-experiment ablation on a dual-encoder retrieval model over 1.3M Yelp reviews, with a leave-one-out onboarding protocol that prevents test-set leakage on cold users.

RESEARCH EXPERIENCE

Lightweight CNN Encoders for Nucleus Segmentation

2025–2026

Sole author · Northeastern University

PyTorch, timm, PanNuke, MoNuSeg

- Designed and ran a controlled encoder ablation isolating encoder choice in U-Net segmentation, holding decoder, augmentation, and optimization schedule fixed across VGG, ResNet, and Swin-T backbones.
- Established that a 34M-parameter VGG U-Net (Dice 0.851 on PanNuke; 0.796 on MoNuSeg) matches a purpose-built medical transformer and is Pareto-dominant on inference cost.
- Attributed the result to spatial-resolution preservation at skip connections and convolutional inductive bias outweighing Swin’s pretraining advantage in the low-data regime.

- Authored the full manuscript, produced all quantitative and qualitative figures, and wrote the discussion.

Two-Tower Recommendation with Cold Start Ablation

2025

Primary contributor & project lead · team of 4

PyTorch, BPR loss, Yelp Open Dataset

- Found context features alone (day-of-week, distance, restaurant temporal profiles) outperform the full model on cold-start users by 4.2 NDCG@10 points, while content features remain essential for cold restaurants — an asymmetry in feature importance across cold-start regimes.
- Designed a leave-one-out onboarding evaluation that prevents information leakage when a cold user's only signal comes from held-out reviews.
- Built a 4-stage data pipeline (extract, preprocess, split, feature prep) with iterative interaction-density filtering and stratified cold-start holdout; reduced dataset construction from 47s to 8s via vectorization.
- Implemented on-device feature indexing eliminating per-batch CPU-GPU transfer, with per-epoch resampling of geographic negatives and Gumbel-max onboarding selections as data augmentation.

Flow Matching for Robotic Manipulation

2025

Team of 3 · Northeastern University

PyTorch, MuJoCo, MetaWorld

- Implemented continuous-time flow matching for imitation learning: a neural vector field transports Gaussian noise to expert action distributions via RK2 ODE integration, avoiding the iterative denoising cost of diffusion policies.
- Scaled from a proprioception-only baseline on single-object bin-picking to an RGBD- and language-conditioned policy with 8-step action chunking across 3-object scenes; characterized compounding-error limits on multi-object generalization.

RailMax: Amtrak Service Optimizer

2025

Open-source modeling study

InterpretML, GTFS, Census ACS, Folium

- Modeled the US Amtrak network as a directed weighted graph and engineered node-level features including destination-population-weighted reachability and eigenvector centrality.
- Trained an Explainable Boosting Machine ($R^2 = 0.76$, RMSE 0.84 log scale) under 5-fold stratified group cross-validation with k-means geographic clustering to prevent spatial leakage.
- Integrated Census ACS, GTFS, NTAD, and IPEDS sources; deployed an interactive what-if frequency simulator identifying the top-20 city pairs most likely to benefit from added service.

PROFESSIONAL EXPERIENCE

Teradata — AI/ML Product Management Analytics Intern

June 2025 – May 2026

San Diego, CA (Remote)

- Authored the business case and led the design of a context-aware AI Notebook Assistant for Vantage-Cloud AI Studio, enabling intelligent code generation, contextual suggestions, and integration with company libraries and documentation.
- Built frontend prototypes and interaction specs for agentic notebook assistant features, translating LLM capability constraints into user-facing product decisions.

PlateMate — Co-Founder & Chief Technology Officer

May 2025 – Present

Boston, MA

- Lead ML architecture and engineering for a food recommendation platform: hybrid item-based collaborative filtering plus semantic vector search over restaurant reviews.

- Built a RAG retrieval pipeline with 768-dimensional `all-mpnet-base-v2` embeddings on MongoDB Atlas Vector Search, parsing natural-language dish queries into semantic and structured geo-aware filters.

LineVision Inc. — Software Engineer Co-op
Boston, MA

Jan 2024 – June 2024

- Full-stack owner of an internal web application; defined features with stakeholders, expanded APIs and database schema, maintained the application end to end.
- Re-architected a data pipeline for staff scientists, achieving a 1000× speed improvement.

Breckinridge Capital Advisors — Infrastructure Intern
Boston, MA

July 2023 – Sept 2023

- Cleaned datasets, built Python web scrapers, and trained a clustering model on the resulting data.
- Performed Windows and Linux system administration and security work using PowerShell, Darktrace, and Bit9; built front-ends in React.js and Razor Pages.

Generate Product Development
Boston, MA

Jan 2023 – May 2025

Project Lead

Dec 2024 – May 2025

- Led 14 engineers and designers to deliver a production mobile application; defined requirements with external stakeholders.

Technical Lead

Sept 2023 – Dec 2023

- Led a team of developers building an iOS/Android app on a React Native frontend with a Go/Postgres backend.

Software Engineer

Jan 2023 – May 2024

- Backend developer on a team of 8 building ReMo, a literacy tool for educators; designed the database and internal model and implemented the networking API in Go.

TEACHING EXPERIENCE

Khoury College of Computer Sciences — Lead Teaching Assistant
Fundamentals of Computer Science I & II

2022 – 2025
Boston, MA

- Led weekly two-hour laboratory sections of 40+ students across six semesters, held 1-on-1 office hours, and graded assignments.
- Taught recursion, linked lists, and functional programming (Fundamentals I), then Java object-oriented design, polymorphism, data structures, algorithm design, and time complexity (Fundamentals II).

HONORS & AWARDS

Rookie of the Year, HackBeanpot

February 2023

Built a food-desert tracking web application on a six-person team.

Students of the Year Runner-Up, Leukemia & Lymphoma Society

March 2021

Raised \$58,000 for blood cancer research in seven weeks during the height of the COVID-19 pandemic.

Eagle Scout, Scouts BSA

October 2019

Led 30 volunteers to design, fundraise, and construct six portable benches for a local school, delivered on schedule despite pandemic restrictions and supply shortages.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript/JavaScript, SQL

Deep Learning: PyTorch, timm, CNNs, U-Net, Vision Transformers (Swin), Transformers & attention, flow matching, diffusion policies, imitation learning

Machine Learning: scikit-learn, InterpretML (EBM), gradient boosting, PCA, SVM, collaborative filtering, two-tower retrieval, ablation design, cross-validation methodology

LLM & Agentic Systems: Claude and OpenAI APIs, MCP servers, multi-agent orchestration, RAG, SentenceTransformers, vector search, NeMo Guardrails, OpenTelemetry

Infrastructure: Docker, AWS (S3), FastAPI, MongoDB Atlas, SLURM / HPC clusters, Git

Data & Visualization: pandas, NumPy, Folium, Streamlit, GTFS, Census ACS

LANGUAGES & ACTIVITIES

Languages: Latin (limited working); English (native).

Activities: Generate Product Development (Northeastern student-run product studio); captain, Northeastern FC (BSSL) and intramural soccer; guitarist and songwriter, Northeastern Jazz Combo and Music Farm.

References available upon request.